9.1 Quantum Oracle

A Boolean function is a function of the form:

$$f : \{0,1\}^n \to \{0,1\},$$

Boolean Function of **one bit**: $f : \{0,1\} \longrightarrow \{0,1\}$,

Example 1: Write the truth tale of $f(x) = \bar{x}$, where is $\bar{x}$ is NOT of x.

x	$f(x)$
0	1
1	0

9.1 Quantum Oracle

$f : \{0,1\}^2 \to \{0,1\}$

Example 2: Write the truth tale of $f(x_0, x_1) = x_0 \wedge x_1$, where is $\bar{x}$ is NOT of x.

x_0	x_1	$f(x_0, x_1)$
0	0	0
0	1	0
1	0	0
1	1	1

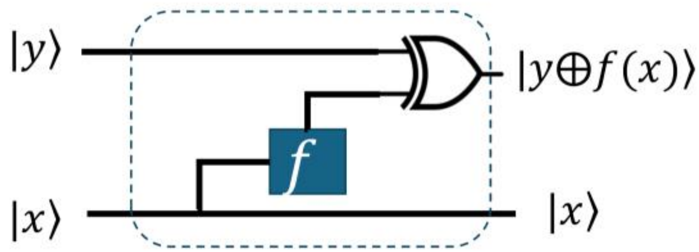
$|x\rangle|y \oplus f(x)\rangle$

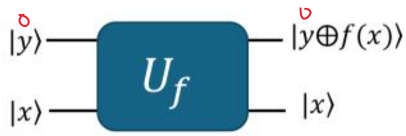
x_0	x_1	y	$f(x_0, x_1)$ ($x_0 \wedge x_1$)	$f(x_0, x_1) \oplus y$	$\lvert x\rangle\lvert y\oplus f(x)\rangle$ ($x_0 x_1$)
0	0	0	0	0	$\lvert 0\,0\rangle 0\rangle$
0	0	1	0	1	0 0 1
0	1	0	0	0	0 1 0
0	1	1	0	1	0 1 1
1	0	0	0	0	1 0 0
1	0	1	0	1	1 0 1
1	1	0	1	1	1 1 1
1	1	1	1	0	1 1 0

9.1 Quantum Oracle

Quantum oracle: is a black box.

Quantum gates are unitary operations which are invertible.





$|x\rangle|f(x)\rangle$

$|x\rangle\, y \oplus f(x)\rangle$

9.1 Quantum Oracle

Quantum Oracles for Functions of **Two Boolean** Inputs:

$$f: \{0,1\}^2 \longrightarrow \{0,1\}$$

$$|x_1, x_2, y\rangle \xrightarrow{U_f} |x_1, x_2, y \oplus f(x_1, x_2)\rangle$$

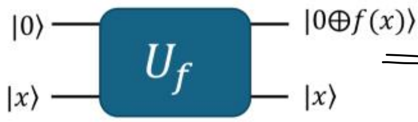
Quantum Oracles for Functions of **Boolean** Inputs:

$$f: \{0,1\}^n \longrightarrow \{0,1\}^m$$

$$|x\rangle_n |y\rangle_m \xrightarrow{U_f} |x\rangle_n |y \oplus f(x)\rangle_m$$

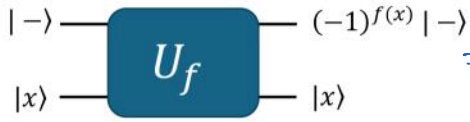
9.1 Quantum Oracle

Quantum Oracle: for fix $y = 0$

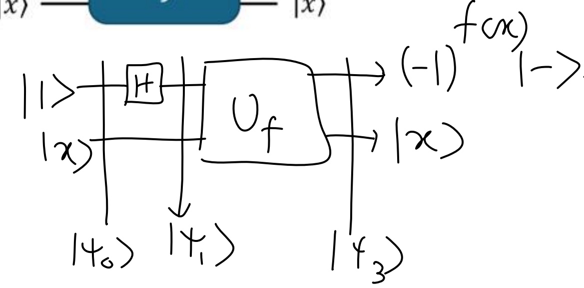


$\Longrightarrow |x\rangle\,|f(x)\rangle$

Phase Oracle:



$\Longrightarrow (-1)^{f(x)}|x\rangle|-\rangle$



① $|\psi_0\rangle = |x\rangle|1\rangle$

② $|\psi_1\rangle = I \otimes H|\psi_0\rangle = |x\rangle|-\rangle = \frac{1}{\sqrt{2}}\left(|x\rangle|0\rangle - |x\rangle|1\rangle\right)$

③ $|\psi_2\rangle = U\,|\psi_1\rangle = \frac{1}{\sqrt{2}}\left(|x\rangle|0 \oplus f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle\right)$

$$= \begin{cases} |x\rangle|-\rangle & \text{If } f(x) = 0 \\ -|x\rangle|-\rangle & f(x) = 1 \end{cases}$$

$$\boxed{= (-1)^{f(x)}\,|x\rangle|-\rangle}$$

9.2 Deutsch's problem and algorithm

Consider a Boolean function $f: \{0,1\} \rightarrow \{0,1\}$,

How many possible values for $f(x)$ for all $x \in \{0,1\}$?

x	f_1	f_2	f_3	f_4
0	0	1	0	1
1	0	1	1	0

$\underbrace{f_1 \quad f_2}_{\text{Constant}}$ $\underbrace{f_3 \quad f_4}_{\text{Balanced}}$

$f(0) = f(1)$ Constant

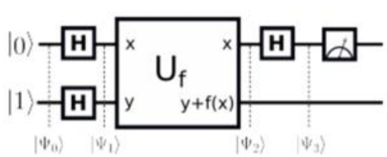
$f(0) \neq f(1)$ balanced.

Classically: How many time to call the function to figure out if the function is constant or balanced?

2 time to call $f(x)$

9.2 Deutsch's problem and algorithm

Deutsch designs a quantum circuit that uses interference and a *single* function call to determine whether the function $f: \{0,1\} \rightarrow \{0,1\}$ is constant or balanced:



① $|\psi_0\rangle = |0\rangle|1\rangle$

② $|\psi_1\rangle = H^{\otimes 2}|01\rangle = |+\rangle|-\rangle$

③ $|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{\sqrt{2}}\left(U_f|0\rangle|-\rangle + U_f|1\rangle|-\rangle\right)$

$|x\rangle|-\rangle \rightarrow \boxed{U_f} \rightarrow (-1)^{f(x)}|-\rangle$ $= \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right)|-\rangle$

$= (-1)^{f(0)}\frac{1}{\sqrt{2}}\left(|0\rangle + (-1)^{f(1)-f(0)}|1\rangle\right)|-\rangle$

$$= \begin{cases} |+\rangle|-\rangle & \text{If } f(0)=f(1) \\ |-\rangle|-\rangle & \text{If } f(0)\neq f(1) \end{cases}$$

④ $|\psi_3\rangle = H \otimes I\,|\psi_2\rangle = \begin{cases} |0\rangle|-\rangle & \text{If } f \text{ constant} \\ |1\rangle|-\rangle & \text{If } f \text{ is balanced.} \end{cases}$

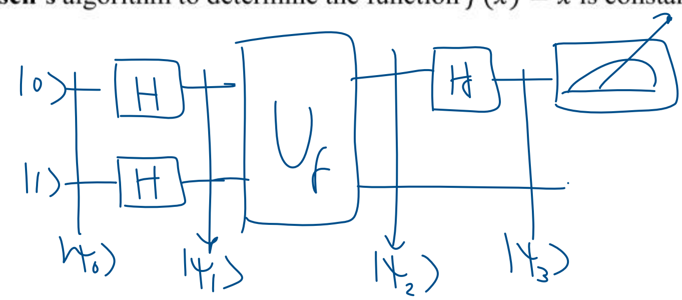
⑤ Measuring the first register

If output $0 \Rightarrow f(x)$ is constant

If output $1 \Rightarrow f(x)$ is balanced.

9.2 Deutsch's problem and algorithm

Example 3: Use **Deutsch's** algorithm to determine the function $f(x) = x$ is constant or balanced?



① $|\psi_0\rangle = |01\rangle$

② $|\psi_1\rangle = H \otimes H\,|01\rangle = |+-\rangle$ $\quad =0 \quad =1$

③ $|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right)|-\rangle$

$= \frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right)|-\rangle = |-\rangle|-\rangle$

④ $|\psi_3\rangle = H \otimes I\,|\psi_2\rangle = |1\rangle|-\rangle$

⑤ Measure First register $\Rightarrow$ output 1

Then $f(x)$ is balanced

9.3 The N-bit Hadamard transformation

- The Hadamard operation applied to many or all qubits is a common element of many quantum algorithms.
- We consider a system of $N = 2^n$ qubits.
- The n −bit Hadamard transformation which is the tensor product of a Hadamard transformation for each bit.

9.3 The n −bit Hadamard transformation

Walsh-Hadamard transformation

$$W_n = H \oplus H \oplus \ldots . \oplus H = H^{\oplus n}$$

Example 4: Find W_3 on 3 qubits starting with $|\psi\rangle = |000\rangle$

$$H^{\otimes 3}|000\rangle = \frac{1}{\sqrt{2^3}} \sum_{x=0}^{7} |x\rangle$$

9.3 The n −bit Hadamard transformation

Walsh-Hadamard transformation for n −qubit $|\psi\rangle = |00 \ldots 0\rangle$

$$W_n = H^{\oplus n}|\psi\rangle$$

$$N = 2^n \qquad = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

Example 5: Find W_3 on 3 qubits starting with $|\psi\rangle = |001\rangle$

$$H^{\otimes}|001\rangle = \frac{1}{\sqrt{2^3}}\Big((|0\rangle+|1\rangle)(|0\rangle+|1\rangle)(|0\rangle-|1\rangle)$$

$|00\rangle+|01\rangle+|10\rangle+|11\rangle$

$$|x\rangle = |x_0 x_1 x_2\rangle \qquad = \frac{1}{\sqrt{2^3}}\big(|000\rangle - |001\rangle + |010\rangle - |011\rangle$$
$$|100\rangle - |101\rangle + |110\rangle - |111\rangle\big)$$

$$= \frac{1}{\sqrt{2^3}} \sum_{x=0}^{7} (-1)^{x_2} |x\rangle$$

9.3 The n −bit Hadamard transformation

Example 6: Find W_3 on 3 qubits starting with $|\psi\rangle = |011\rangle$

$$H^{\otimes}|011\rangle = \frac{1}{\sqrt{2^3}}\big(|0\rangle+|1\rangle\big)\big(|00\rangle - |01\rangle - |10\rangle + |11\rangle\big)$$

$x_1 x_2$

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle \qquad = \frac{1}{\sqrt{2^3}}\big(|000\rangle - |001\rangle - |010\rangle + |011\rangle$$
$$+|100\rangle - |101\rangle - |110\rangle + |111\rangle\big) = \frac{1}{\sqrt{2^3}}\sum_{x=0}^{7}(-1)^{x_1+x_2}|x\rangle$$

We can define $x \cdot y$ for $x = (x_0, x_1, \ldots, x_{n-1})$and $y = (y_0, y_1, \ldots, y_{n-1})$are both $n - bit$ strings as follows

$$x \cdot y = x_0 \cdot y_0, +x_1 \cdot y_1 \ldots + x_{n-1} \cdot y_{n-1}\ (mod\ 2) \qquad y_0 y_1 y_2$$

$$H^{\otimes n}|y\rangle \qquad W_n = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{x\cdot y}|x\rangle$$

$$\underline{H^{\otimes}|x\rangle} = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{xy} |y\rangle$$

$$H^{\otimes 3}|011\rangle$$
$$= (-1)^{(000)(011)}$$
$$(-1)^{(001)(011)} \qquad |001\rangle$$

9.4 Generalized Deutsch's Problem

Generalized Deutsch's Problem determine whether the function

$f:\{0,1\}^n \rightarrow \{0,1\}$ is constant or balanced:

Classical Solution:

x	$f(x)$	$f(x)$
$00\cdots0$	0 or 1	half 0 and half 1 } balanced
$00\cdots01$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
$11\cdots1$	0	1

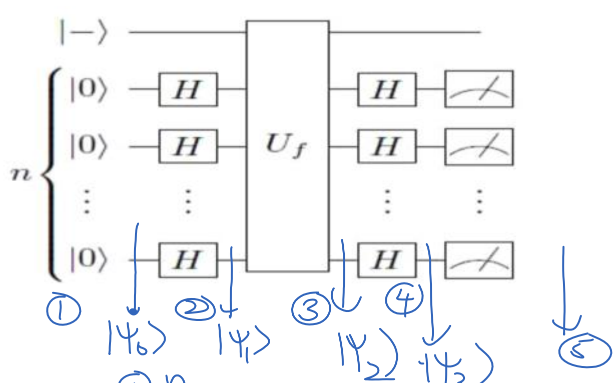
$\underbrace{\quad}_{\text{constant}}$

$$\frac{2^n}{2}+1 = 2^{n-1}+1 \Rightarrow O(2^n)$$

9.4 Generalized Deutsch's Problem

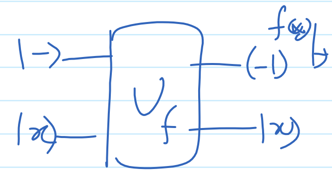
Quantum Solution: Deutsch-Jozsa Algorithm

Deutsch-Jozsa circuit



① $|\psi_0\rangle = |0\rangle^{\otimes n}|-\rangle$

② $|\psi_1\rangle = \left(H^{\otimes n}|0\rangle^{\otimes n}\right)|-\rangle = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle|-\rangle$



③ $|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}U_f|x|-\rangle$

$= \frac{1}{}\sum^{2^n-1}(-1)^{f(x)}|x\rangle|-\rangle$

$$\textcircled{4}\ |\psi_3\rangle = H^{\otimes n} \otimes I\, |\psi_2\rangle$$

$$= \left(\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)}\, H^{\otimes n}|x\rangle\right) |-\rangle \qquad |x_0 x_1 \cdots x_{n-1}\rangle$$

$$= \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y} |y\rangle |-\rangle$$

$$= \frac{1}{2^n} \sum_{x=0}^{2^n-1} \sum_{y=0}^{2^n-1} (-1)^{f(x)+\overbrace{x\cdot y}^{=0}} |y\rangle\, |-\rangle$$

$\textcircled{5}$ Measure the first register $|y\rangle \Longrightarrow |0\rangle$

$\Longrightarrow$ the amplitude will $\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)}$

$\sum_{x=0}^{2^n-1} (-1)^{f(x)}$

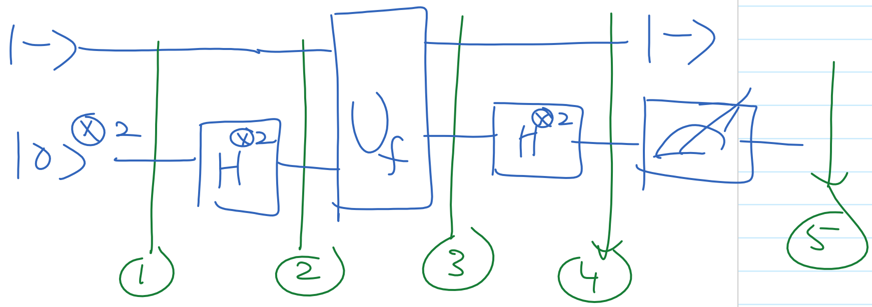
$\sum_{x=0}^{2^n-1} -1 = -1-1-1-1 = -2^n$

$$\text{prob}(0) = \left| \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \right|^2 = \begin{cases} 1 \Rightarrow f(x) \text{ is constant} \\ 0 \Rightarrow f \text{ is balanced.} \end{cases}$$

If the probability not 1 or 0 $\Longrightarrow$ neither constant nor balanced

9.4 Generalized Deutsch's Problem

Example 7: Use Deutsch-Jozsa algorithm to show that $f(x_0, x_1) = 0$ is constant



(1) $|\psi_0\rangle = |0\rangle^{\otimes 2} |-\rangle$

(2) $|\psi_1\rangle = \left(H^{\otimes 2} |0\rangle^{\otimes 2}\right)|-\rangle = \frac{1}{\sqrt{2^2}} \sum_{x=0}^{3} |x\rangle |-\rangle$

(3) $|\psi_2\rangle = U_f |\psi_1\rangle = \frac{1}{2} \sum_{x=0}^{3} (-1)^{f(x)} |x\rangle |-\rangle$

$= \frac{1}{2} \sum_{x=0}^{3} |x\rangle |-\rangle$

(3) $|+\rangle|+\rangle|-\rangle$

(4) $\left(H^{\otimes 2} |+\rangle^{\otimes 2}\right)|-\rangle$

$= |00\rangle|-\rangle$

(4) $|\psi_3\rangle = H^{\otimes n} \otimes I\, |\psi_2\rangle = \frac{1}{2} \sum_{x=0}^{3} \frac{1}{2} \sum_{y=0}^{3} (-1)^{x \cdot y} |y\rangle |-\rangle$

(5) Measure first register to $|0\rangle$

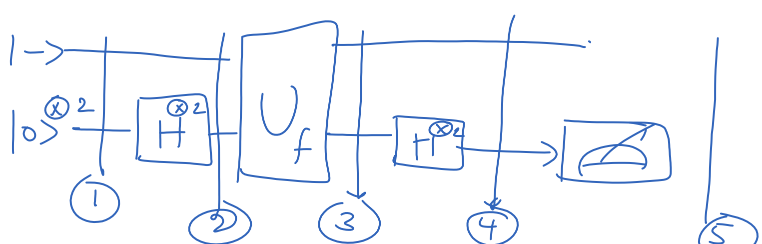
Then, amplitude $\Rightarrow \frac{1}{2^2} \sum_{x=0}^{3} 1 = \frac{4}{4} = 1$

$\sum_{y=0}^{3} 1 = 1 + 1$

$\therefore Prob(0) = |1|^2 = 1$

9.4 Generalized Deutsch's Problem

Example 8: Use Deutsch-Jozsa algorithm to show that $f(x_0, x_1) = x_0 \oplus x_1$ is balanced



(1) $|\psi_0\rangle = |0\rangle^{\otimes 2} |-\rangle$

2 3 4 5

1. $|\psi_0\rangle = |0\rangle^{\otimes 2}|-\rangle$
2. $|\psi_1\rangle = \left(H^{\otimes 2}|0\rangle^{\otimes 2}\right)|-\rangle = \frac{1}{\sqrt{2^2}}\sum_{x=0}^{3}|x\rangle|-\rangle$
3. $|\psi_2\rangle = U_f|\psi_1\rangle = \frac{1}{2}\sum_{x=0}^{3}(-1)^{f(x)}|x\rangle|-\rangle$
4. $|\psi_3\rangle = H^{\otimes 2}\otimes I|\psi_2\rangle = \frac{1}{2}\frac{1}{2}\sum_{x=0}^{3}\sum_{y=0}^{3}(-1)^{f(x)+x\cdot y}\underline{|y\rangle}|-\rangle$
5. Measuring the first register to $|0\rangle$

Amplitude $\Longrightarrow \frac{1}{4}\sum_{x=0}^{3}(-1)^{f(x)}$ $\quad\boxed{f(x_0,x_1) = x_0 \oplus x_1}$

$$= \frac{1}{4}\left\{(-1)^{f(0,0)} + (-1)^{f(0,1)} + (-1)^{f(1,0)} + (-1)^{f(1,1)}\right\}$$

$$= \frac{1}{4}\left\{(1) + (-1) + (-1) + (1)\right\} = \underline{\underline{0}}$$

$\therefore P(0) = 0 \Longrightarrow f$ is balance